

1. Dataset Overview

This analysis is based on transaction records from AmazingMart EU, combining three interconnected sheets:

a. ListOfOrders

Contains customer-level sales information including:

Variable Name	Description
Order ID	Unique identifier for each order
Order Date	Date the order was placed
Customer Name	Name of the customer
City, Country, State, Region	Location details
Segment	Customer type (Consumer, Corporate, Home Office)
Ship Date	Shipment date
Ship Mode	Shipping service used

b. OrderBreakdown

Breaks down individual products in orders:

Variable Name	Description
Product Name	Item purchased
Category	High-level category (Furniture, Technology)
Sub-Category	More detailed classification
Quantity	Units sold
Discount	Discount applied to the item
Sales	Revenue from item
Profit	Net profit from item

c. SalesTargets

Shows monthly sales targets per category.

Variable Name	Description
Month of Order	Month start date
Category	Product category
Target	Monthly sales target (currency)

2. Key Questions and Analytical Goals

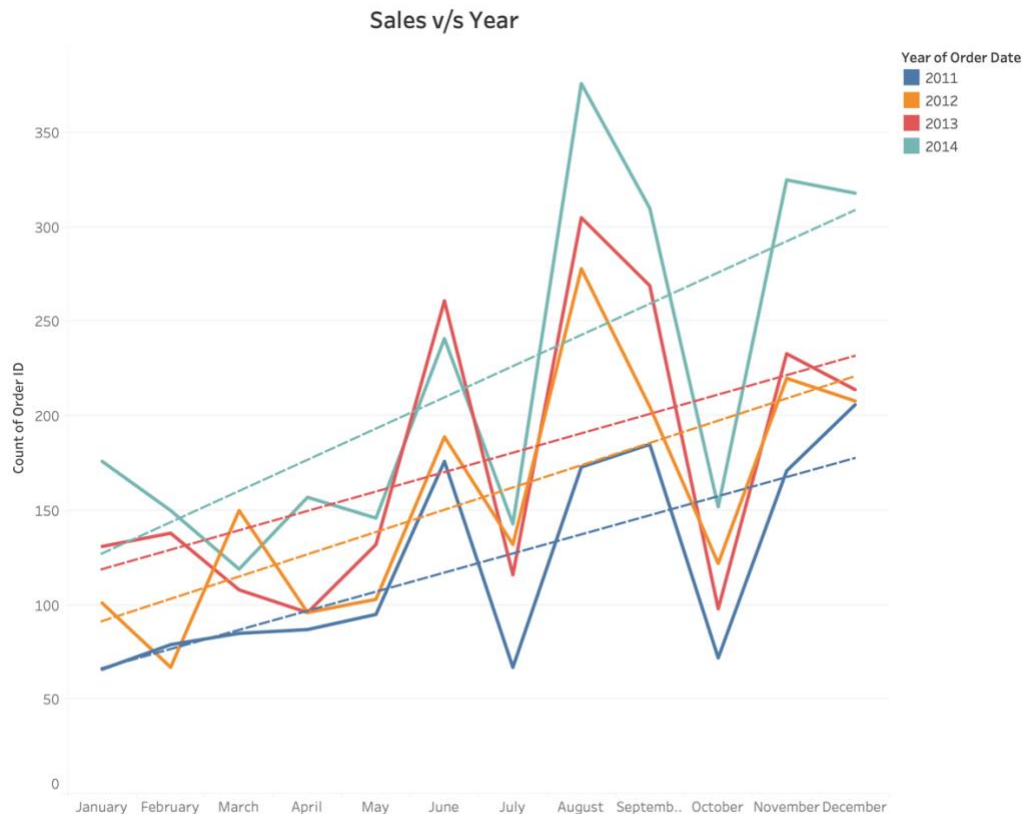
The following business questions guide this analysis:

1. Which product categories and sub-categories generate the most revenue and profit?

2. How do customer segments (Consumer, Corporate, Home Office) vary in terms of performance?
3. Which regions/countries are underperforming or overachieving relative to targets?
4. How does discounting impact profitability?
5. Are monthly sales trends aligned with business targets?

Our goal is to provide strategic insights using visualizations and data summaries and translate findings into recommendations for business improvement.

Visualization 1: Monthly Order Trends by Year



Breakdown: Each line represents a different year from the dataset, with a trendline included per year.

Data Summary:

Month of Order Date	Year of Order Date	Count of Order ID
January	2014	176
February	2014	150
March	2014	119
April	2014	157
May	2014	146
June	2014	241
July	2014	143
August	2014	376
September	2014	310
October	2014	152
November	2014	325
December	2014	318
January	2013	131
February	2013	138
March	2013	108
April	2013	96
May	2013	132
June	2013	261
July	2013	116
August	2013	305
September	2013	269
October	2013	98
November	2013	233
December	2013	214

January	2012	101
February	2012	67
March	2012	150
April	2012	96
May	2012	103
June	2012	189
July	2012	132
August	2012	278
September	2012	205
October	2012	122
November	2012	220
December	2012	208
January	2011	66
February	2011	79
March	2011	85
April	2011	87
May	2011	95
June	2011	176
July	2011	67
August	2011	173
September	2011	185
October	2011	72
November	2011	171
December	2011	206

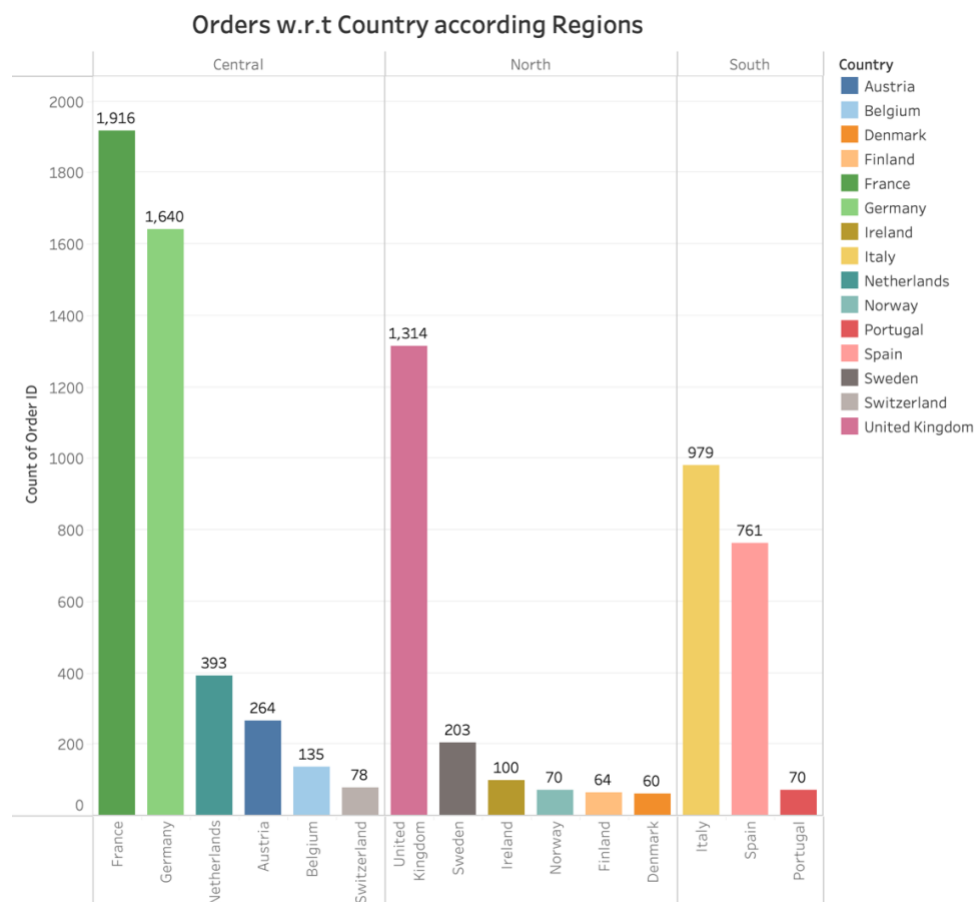
Key Observations:

- There is a steady growth in the number of orders placed each year from January to December.
- August and November consistently exhibit spikes in order counts, suggesting these are peak sales months—potentially influenced by summer and holiday seasons.
- Order volume increased year-over-year, showing business growth and increased customer engagement.
- The trendlines indicate positive momentum in order activity across all years, with the most recent year (green) showing the steepest upward trend.

Business Implications:

- Sales campaigns should be intensified in August and November, possibly by aligning promotions with historical behavior.
- Staffing, inventory, and logistics should be scaled in Q3 and Q4 to manage the rising order volumes efficiently.
- Trendline analysis supports forecasting models that anticipate further growth in future years.

Visualization 2: Orders by Country and Region



Count of Order ID for each Country broken down by Region. Color shows details about Country. The view is filtered on Region, which keeps Central, North and South.

Breakdown: Each bar represents a country, grouped by region (Central, North, South).

Data Summary:

Orders by country and region

Region	Country	Count of Order ID	Distinct count of Order ID
Central	Switzerland	78	40
Central	Belgium	135	68
Central	Austria	264	135
Central	Netherlands	393	194
Central	Germany	1640	806
Central	France	1916	991
North	Denmark	60	29
North	Finland	64	34
North	Norway	70	37
North	Ireland	100	50
North	Sweden	203	100
North	United Kingdom	1314	700
South	Portugal	70	37
South	Spain	761	403
South	Italy	979	493

Key Insights:

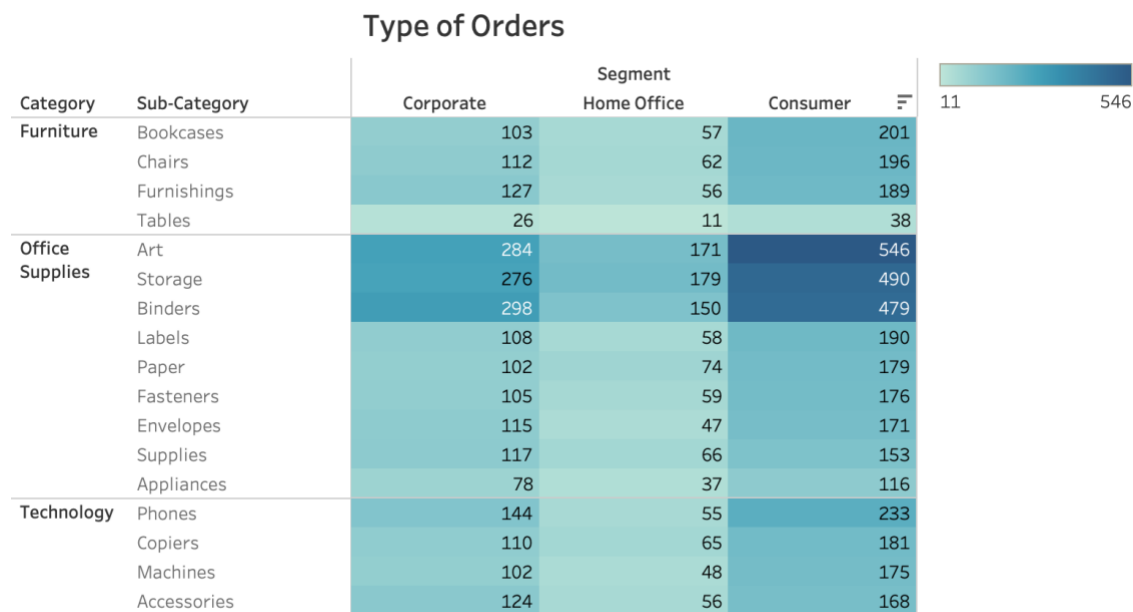
- France (1,916 orders) and Germany (1,640 orders) dominate the Central region, indicating high customer activity.
- In the North, the United Kingdom is the top performer (1,314 orders), significantly ahead of Sweden (203) and Ireland (100).

- Italy (979) and Spain (761) lead in the Southern region, contributing strong order counts relative to their regional peers.
- Countries like Belgium, Switzerland, Portugal, Denmark, and Norway show very low activity, suggesting either market gaps or untapped potential.

Business Implications:

- Marketing and logistics investments should prioritize France, Germany, UK, Italy, and Spain, as they show the highest demand.
- Countries with low order counts should be analyzed further:
 - a) Are there distribution issues?
 - b) Is local marketing insufficient?
 - c) Are pricing or localization barriers present?
- Launch region-specific campaigns in underperforming countries to assess response before larger investments.

Visualization 3: Order Types by Segment and Sub-Category



Distinct count of Order ID broken down by Segment vs. Category and Sub-Category. Color shows distinct count of Order ID. The marks are labeled by distinct count of Order ID.

Breakdown: Matrix showing sub-category order volume across Corporate, Home Office, and Consumer segments.

Data Summary:

Order type by segment and category

Category	Sub-Category	Segment	Distinct count of Order ID
Furniture	Tables	Consumer	38
Furniture	Furnishings	Consumer	189
Furniture	Chairs	Consumer	196
Furniture	Bookcases	Consumer	201
Furniture	Tables	Home Office	11
Furniture	Furnishings	Home Office	56
Furniture	Chairs	Home Office	62
Furniture	Bookcases	Home Office	57
Furniture	Tables	Corporate	26
Furniture	Furnishings	Corporate	127
Furniture	Chairs	Corporate	112
Furniture	Bookcases	Corporate	103
Technology	Accessories	Consumer	168
Technology	Machines	Consumer	175
Technology	Copiers	Consumer	181
Technology	Phones	Consumer	233
Technology	Accessories	Home Office	56
Technology	Machines	Home Office	48
Technology	Copiers	Home Office	65
Technology	Phones	Home Office	55
Technology	Accessories	Corporate	124
Technology	Machines	Corporate	102
Technology	Copiers	Corporate	110
Technology	Phones	Corporate	144

Office Supplies	Appliances	Consumer	116
Office Supplies	Supplies	Consumer	153
Office Supplies	Envelopes	Consumer	171
Office Supplies	Fasteners	Consumer	176
Office Supplies	Paper	Consumer	179
Office Supplies	Labels	Consumer	190
Office Supplies	Binders	Consumer	479
Office Supplies	Storage	Consumer	490
Office Supplies	Art	Consumer	546
Office Supplies	Appliances	Home Office	37
Office Supplies	Supplies	Home Office	66
Office Supplies	Envelopes	Home Office	47
Office Supplies	Fasteners	Home Office	59
Office Supplies	Paper	Home Office	74
Office Supplies	Labels	Home Office	58
Office Supplies	Binders	Home Office	150
Office Supplies	Storage	Home Office	179
Office Supplies	Art	Home Office	171
Office Supplies	Appliances	Corporate	78
Office Supplies	Supplies	Corporate	117
Office Supplies	Envelopes	Corporate	115
Office Supplies	Fasteners	Corporate	105
Office Supplies	Paper	Corporate	102
Office Supplies	Labels	Corporate	108
Office Supplies	Binders	Corporate	298
Office Supplies	Storage	Corporate	276
Office Supplies	Art	Corporate	284

Key Insights:

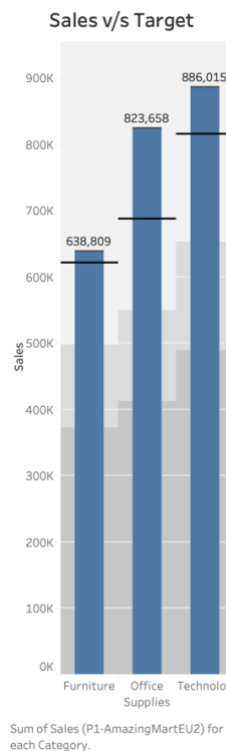
- The Consumer segment is the most active, especially in Office Supplies, with standout performance in:
 - a) Art (546 orders)
 - b) Storage (490 orders)

- c) Binders (479 orders)
- Corporate segment places high volume orders for:
 - a) Binders (298), Art (284), Storage (276)
- Technology products such as Phones, Copiers, and Machines are consistently ordered across all segments but in lower volumes.
- Furniture sub-categories like Bookcases, Chairs, and Furnishings are mostly favored by Consumers, with limited adoption in Corporate or Home Office segments.

Business Implications:

- Office Supplies are the dominant category across all segments — tailor product bundles, promotions, or subscriptions around these products, especially for consumers and corporates.
- Consumer behavior indicates strong demand for creative/stationery-related products like Art and Binders — opportunities exist for seasonal marketing, e.g., back-to-school, academic kits, etc.
- Corporate and Home Office segments may benefit from bulk purchase incentives, given their tendency to order practical supplies like storage and binders.
- For Furniture and Technology categories, more aggressive B2B outreach may help boost segment diversification beyond Consumer users.

Visualization 4: Sales vs Target by Category



Metric Displayed: Total Sales per category with overlaid Target values (black lines).

Category	Actual Sales	Target	Performance
Furniture	638,809	600,000	Target Met
Office Supplies	823,658	700,000	Target Exceeded
Technology	886,015	800,000	Target Exceeded

Data Summary:

Sales vs Target by Category

Category	Sales	Target
Technology	886015	816000
Office Supplies	823658	687000
Furniture	638809	621600

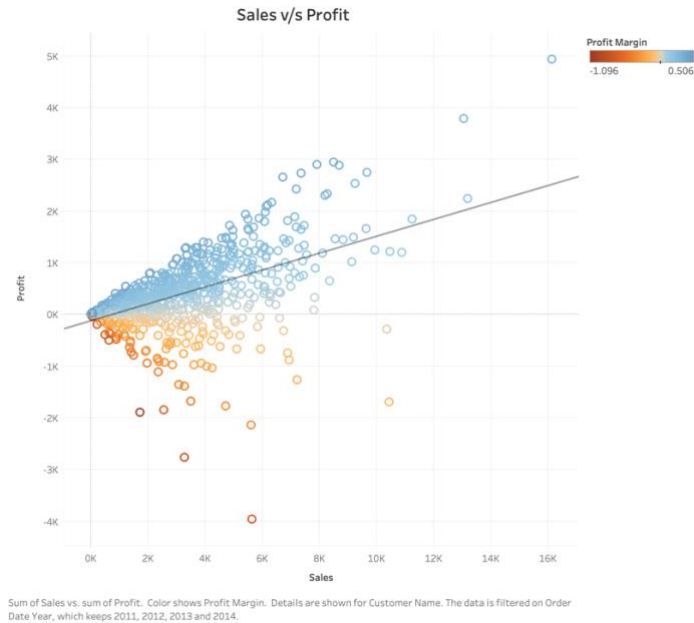
Key Insights:

- All three product categories met or exceeded their sales targets, showcasing strong business performance.
- Technology achieved the highest sales overall and exceeded its target by a significant margin.
- Office Supplies also demonstrated robust performance, with a ~120K surplus over its goal.
- Furniture, while meeting targets, continues to underperform relative to other categories and shows the narrowest margin.

Business Implications:

- Technology and Office Supplies should receive more focus for upselling and new product launches, as demand already aligns with strategic goals.
- Furniture, despite hitting its target, may require:
 - a) Product line optimization
 - b) Pricing or promotional revamps
 - c) Inventory reviews to improve profitability (especially based on earlier sub-category loss trends)

Visualization 5: Sales vs. Profit (Customer-Level Analysis)



Metric Displayed: Scatter plot showing total Sales vs. Profit by customer, with a linear trendline and color-coded profit margins.

Data Summary:

Sales vs Profit

Customer Name	Profit Margin	Profit	Sales
Zona Meyer	0.226120359	883	3905
Zoe Lowin	0.159064994	279	1754
Zoe Lanigan	-0.232379194	-755	3249
Zara Money	0.335429769	960	2862
Zara Loton	0.272614622	440	1614
Zara Jennings	0.269662921	264	979
Zachary Warner	0.115942029	584	5037
Zachary Disher	0.306661081	1464	4774
Yasmin Higgins	0.038461538	14	364
Yasmin Bradley	0.193351801	349	1805
Xavier Synan	0.233367983	898	3848

Key Observations:

- The overall trend is positive, indicating that higher sales generally lead to higher profits.

- However, a large cluster of customers generates negative profit (orange bubbles) — especially those with mid-to-high sales between \$2K–\$6K.
- These negative-profit transactions may be due to:
 - Excessive discounting
 - High shipping or fulfillment costs
 - Low-margin products
- High-margin customers (blue) not only bring in more revenue but also show long-term profitability potential.

Business Implications:

- Perform customer segmentation based on profitability and tailor loyalty programs or offers for the top-tier profitable ones.
- Investigate customers with high sales but negative margins — possibly due to loss-leader products or poorly structured discounting.
- Limit bulk discounts for low-margin items unless they trigger volume thresholds to cover costs.
- Build a profit-focused sales strategy rather than just revenue growth.